Online Appendix to: Frequent Losers as a Cartel Screen: Detecting Cover Bidding Without Bid Microdata

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Abstract

This online appendix accompanies *Frequent Losers as a Cartel Screen: Detecting Cover Bidding Without Bid Microdata* and contains: (A) the cover-bidding framework and proofs of the three formal results invoked in the main text; (B) core supporting tables and figures; (C) identification audits (Oster bounds and the anti-leakage audit); (D) threshold and specification robustness tables; (E) operational metrics and detector comparison; and (F) the staggered-design estimators that return null at the available bandwidths. References to sections, equations, and tables in the main text use the same labels as the published manuscript.

Appendix A. Cover-Bidding Framework and Proofs

This appendix formalizes the separating-equilibrium framework referenced throughout the main text. Three results are derived and used: a lemma establishing the bright-line filter wins = 0 as the equilibrium choice of the cover-bidder type (Lemma 1); a proposition identifying $\log(1 + \text{tenders_count})$ as a sufficient ranking statistic for the cover-bidder type given award-record data (Proposition 2); and a proposition characterizing the comparative static of optimal cover-bidder deployment in the procuring unit's principal cost of detection (Proposition 3). The framework is an organizing device for the empirical patterns; it is not the source of identification.

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Setup. A cartel of n firms in a procurement market controls a designated winner with bid b^* and deploys $m \geq 0$ cover bidders, each submitting a bid $b_\ell > b^*$. Per-bidder deployment cost is $c_1 > 0$; the cartel faces a detection probability $\theta_k \in (0, 1)$ that depends on oversight capacity in market k (smaller, less-supervised buyers have lower θ_k). The cartel maximizes expected surplus $\pi(m, \theta_k, c_1)$, the difference between cartel rents and the sum of cover-bidder costs and expected detection penalties. We impose two standard assumptions:

A1 (Profit positivity). $\partial\pi/\partial m\big|_{m=0} > 0$: the first cover bidder generates positive marginal cartel surplus, otherwise no cartel deploys cover.

A2 (Diminishing returns). $\partial^2\pi/\partial m^2 < 0$: the surplus function is strictly concave in m, so an interior optimum exists.

The empirical environment is described by award-record data: each firm i has a participation count tenders_count_i, a win indicator $\text{wins}_i \in \{0, 1\}$, and a tenure τ_i. The construct flags firms with $\text{wins}_i = 0$ and tenders_count_i above the operational threshold 14. Lemma 1 below formalizes the role of the wins-zero condition; Proposition 2 formalizes the role of the participation-count ranking.

Lemma 1 (Wins-zero separating equilibrium). *Suppose cover bidders draw their bid b_C from a strategy that satisfies (a) $b_C > b^*$ in equilibrium (cartel-allocation constraint) and (b) winning triggers either capacity revelation (undermining the cover identity in subsequent rounds) or internal cartel sanctions on the deviating firm. Genuine entrants of type G face no such constraint and accept any $b_G < b^*$ that yields positive expected payoff. Then in the separating equilibrium, cover bidders satisfy $\Pr(win \mid C) = 0$: the bright-line filter $wins = 0$ is the equilibrium choice of type C.*

Proof of Lemma 1. Write the cover-bidder per-period payoff as $u_C(b) = -c_1 - \theta_k\phi_0 \cdot \mathbf{1}[\text{win}] - \kappa \cdot \mathbf{1}[\text{win}]$, where $\phi_0 > 0$ is the marginal expected detection penalty per cover bid and $\kappa > 0$ captures expected cartel-internal sanction or capacity-revelation cost. Choosing $b_C > b^*$ delivers $\Pr(\text{win} \mid b_C, b^*) = 0$ in any equilibrium where the designated winner submits b^*, so $u_C = -c_1$. Any deviation to $b'_C < b^*$ delivers $u'_C = -c_1 - \theta_k\phi_0 - \kappa + \text{rents}$, where rents is the deviation gain from winning. Under the cartel's allocation rule, rents $<$

κ for any cover bidder (otherwise the allocation would not be incentive-compatible). Hence $u_C > u'_C$, and the equilibrium strategy is $b_C > b^*$ with $\Pr(\text{win} \mid C) = 0$. By contrast, type G has no κ penalty and accepts $b_G < b^*$ when expected payoff is positive. Hence the empirical filter wins $= 0$ identifies the type-C subset, conditional on the cartel deploying its allocation rule. □ □

Proposition 2 (Linearized cumulative exposure). *Suppose firm i is a cover bidder of type C deployed by a cartel in market k at per-period rate $\lambda_C(\theta_k, c_1)$ and tenure τ_i. Under A1–A2 and a stationary deployment process,*

$$\mathbb{E}[\, \textit{tenders_count}_i \mid \textit{type} = C,\, \tau_i] \;=\; \lambda_C(\theta_k, c_1)\, \tau_i.$$

For genuine competitive entrants of type G, $\mathbb{E}[\textit{tenders_count}_i \mid \textit{type} = G, \tau_i] = \lambda_G\, \tau_i$ with $\lambda_G < \lambda_C$ in any cartel-targeted market: by Proposition 3 below, $m^* > 0$ implies positive equilibrium deployment of cover bidders, while genuine entry does not respond to cartel demand for participation slots. The log-statistic $\log(1+\textit{tenders_count}_i)$ is therefore monotone in λ given tenure, and—under monotone likelihood ratio in the Poisson approximation $T_i \sim \text{Poisson}(\lambda\tau_i)$—the posterior $\Pr(\textit{type} = C \mid \textit{tenders_count}_i = t)$ is non-decreasing in t. Combined with Lemma 1, the joint statistic $(\log(1 + \textit{tenders_count}), \mathbf{1}[\textit{wins} = 0])$ is a sufficient screen for the cover-bidder type given award-record data; the deployable binary frequent-loser rule is the information-coarsening of this latent posterior under the separating equilibrium.

Proof of Proposition 2. The expectation result is immediate from the law of large numbers applied to the deployment counting process given constant rate λ. The monotonicity claim follows from Bayes' rule:

$$\Pr(C \mid t) = \frac{\Pr(t \mid C)\, \pi_C}{\Pr(t \mid C)\, \pi_C + \Pr(t \mid G)\, (1 - \pi_C)},$$

which is non-decreasing in t if and only if the likelihood ratio $\Pr(t \mid C)/\Pr(t \mid G)$ is non-decreasing in t. For Poisson likelihoods with rates $\lambda_C > \lambda_G$, this monotone-likelihood-ratio property is standard. Hence $\log(1+\text{tenders_count})$ is a sufficient ranking statistic for the cover-bidder type, and the binary indicator $\mathbf{1}[\text{tenders_count} > k]$ is its information-coarsening operational form. □ □

Proposition 3 (Optimal cover-bidder deployment). *Under A1–A2, the cartel's optimal cover-bidder deployment rate m^* satisfies the first-order condition $\partial\pi/\partial m =$*

$c_1 + \theta_k \phi_0$, where $\phi_0 > 0$ is the marginal expected detection penalty per cover bidder evaluated at the optimum. The implicit comparative statics are

$$\frac{\partial m^*}{\partial \theta_k} = -\frac{\phi_0}{\partial^2 \pi / \partial m^2} < 0, \qquad \frac{\partial m^*}{\partial c_1} = -\frac{1}{\partial^2 \pi / \partial m^2} < 0.$$

Proof of Proposition 3. The first-order condition follows directly from differentiating $\pi(m, \theta_k, c_1)$ with respect to m and equating the marginal cartel surplus to the marginal cost of the cover bidder (c_1 deployment cost plus expected detection penalty $\theta_k \phi_0$). By A2, $\partial^2 \pi / \partial m^2 < 0$, so the implicit interior solution $\tilde{m}$ defined by the FOC is unique. Implicit differentiation of the FOC yields the comparative statics displayed, both negative because the denominator is negative. The empirical implication is that cover-bidder deployment is more aggressive where the procuring unit's principal cost of detection is lower (§7.2). □ □

Appendix B. Core Supporting Tables and Figures

Table B.1: Tender Counts by Procurement Modality and Year

Year	Pregão	Convite	Total
2009	137,424	232,765	370,189
2010	134,181	253,196	387,377
2011	72,894	140,305	213,199
2012	129,664	305,808	435,472
2013	125,324	328,616	453,940
2014	118,274	310,947	429,221
2015	92,324	226,503	318,827
2016	57,804	122,841	180,645
2017	120,194	261,205	381,399
2018	60,518	121,698	182,216
2019	11,857	26,976	38,833

Notes: Winner-level observations from BEC (São Paulo) 2009–2019. Pregão = electronic reverse auction; Convite = sealed-bid invitation.

Table B.2: Conditional Descriptive Statistics: Residualized Prices by FL Status

	FL-Present (losers = 1)	FL-Absent (losers = 0)
Panel A: Unconditional		
Mean log(price)	4.9241	2.4938
Difference	2.4303	
Panel B: Conditional on Item + Year + PBU FE		
Mean residualized price	0.0472	-0.0024
SD residualized price	1.5734	0.8704
Difference (gap)	0.0495	
Overlap coefficient	0.978	
t-statistic	8.81	
p-value	0.000000	
Cohen's d	0.0390	
Observations	79,452	1,574,949

Notes: Residuals from regressing log(negotiated price) on item, year, and purchasing unit (PBU) fixed effects. The gap represents the mean difference in residualized prices between FL-present and FL-absent tenders, after absorbing common item-level, temporal, and PBU-level variation. Overlap coefficient = $2\Phi(-|\Delta\mu|/\sqrt{\sigma_1^2 + \sigma_0^2})$ measures the proportion of the two residual distributions that overlap; values near 1 indicate near-identical distributions. Cohen's d measures the standardized effect size.

Table B.3: FL Price Effect by Oversight Proxy

Sub-sample	Coefficient	SE	N
By PBU size quartile:			
Q1 (lower oversight)	0.2138	(0.1670)	7,503
Q2 (lower oversight)	0.0111	(0.0513)	87,571
Q3 (higher oversight)	0.0660**	(0.0304)	302,258
Q4 (higher oversight)	0.0165	(0.0160)	1,257,069
By procedure type:			
Pregão (electronic, more transparent)	0.0933***	(0.0255)	546,549
Convite (sealed-bid, less transparent)	0.0382**	(0.0186)	1,107,852

Notes: If FL coefficient is smaller under higher oversight, this is consistent with the regime model where detection probability θ constrains cartel behavior. Item and year FE (PBU FE for procedure split). SE clustered at item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

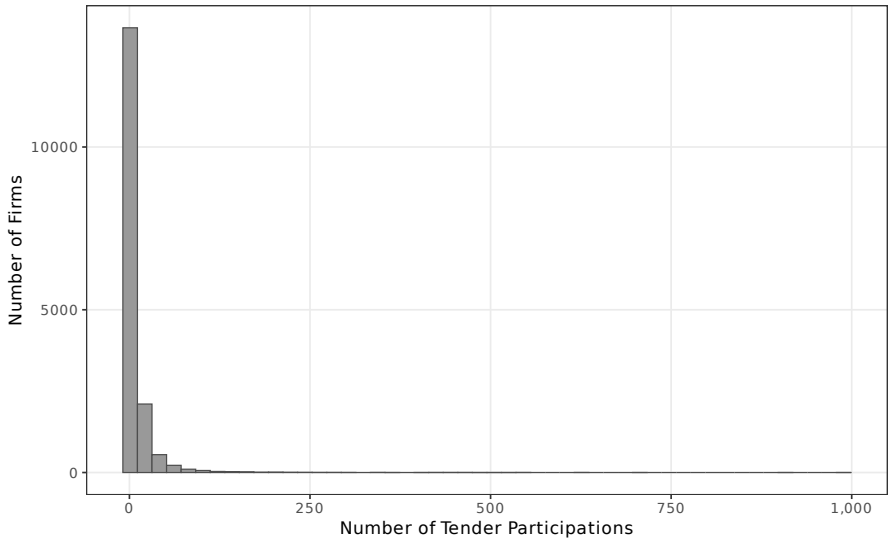

Figure B.1: Distribution of tender-item participation counts across always-loser firms in BEC, 2009–2019. The distribution is heavily right-skewed; a small share of always-losers participates in tens or hundreds of tender-items without ever winning.

Table B.4: Falsification: Loser-Side Concentration in Pregão-Only Subsample (No Minimum-Bidder Rule)

Specification	Pregão-only	Convite-only	Full sample	Ratio (Pregão / Convite)
Binary FL14	$+0.0959^{***}$	$+0.0392^{**}$	$+0.0653^{***}$	2.45×
	(0.0256)	(0.0188)	(0.0216)	
	$p < 10^{-3}$	$p = 0.037$	$p = 0.003$	
Continuous log(1+ max tc)	$+0.0262^{***}$	$+0.0124^{**}$	—	2.11×
	(0.0063)	(0.0049)	—	
	$p < 10^{-4}$	$p = 0.011$	—	
Joint specification: both regressors				
Binary FL14 (with continuous control)	−0.0652	-0.0828^{**}	—	—
	(0.0449)	(0.0350)	—	—
Continuous (with binary control)	$+0.0399^{***}$	$+0.0305^{***}$	—	—
	(0.0118)	(0.0101)	—	—
Items	543,752	1,105,852	1,653,658	—
Item, year, PBU FE	YES	YES	YES	—
Cluster (item)	YES	YES	YES	—
Minimum-bidder rule binds	NO	YES	MIXED	—

Notes: Outcome is $\log p_{\text{negotiated}}$. The pregão-only subsample is the falsification sample for the "regulatory artifact" alternative reading: pregão (Lei 10.520/2002) imposes *no* minimum-bidder rule, so any coordination flagged by the loser-side construct in this subsample cannot be a mechanical consequence of statutory quorum filling. The pregão-only FL premium (+9.6%) is approximately 2.45× larger than the convite-only premium (+3.9%) and strongly significant. Reading: the loser-side concentration construct identifies coordination behavior across modal regimes, not a regulatory artifact specific to convite. The continuous specification confirms this: +2.6% in pregão vs +1.2% in convite. In joint specifications, the binary FL14 flips sign within both modal subsamples (the continuous score absorbs the informational content), consistent with the horse-race result in Table 6; the continuous score is the substantive object, FL14 is the operational implementation. SEs clustered at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: McCrary Density Diagnostic at the Convite/Pregão Statutory Threshold

	Density at threshold			Modality
Threshold (Lei 8.666 art. 23)	Left	Right	Log-density disc.	Convite share gap
R$80,000 (pre-2018)	381	406	−0.063	0.098
R$176,000 (post-2017)	216	277	−0.250	0.092

Notes: Densities estimated by local linear smoothing of contract-record counts in ±50% windows around each statutory cap. The convite-share gap is the difference between the share of contracts using the convite modality just below versus just above the threshold. Both thresholds show small log-density discontinuities ($|\cdot| < 0.25$); neither shows the kind of bunching that would indicate strategic gaming of the modality assignment by buyers. The ~ 9.8 percentage-point convite-share drop across the cap is consistent with rule-driven assignment of value to modality, not with strategic selection.

Table B.6: Price Regressions Excluding CADE-Involved Markets

	(1) General+PBU	(2) Pregão
FL presence	0.0616***	0.0901***
	(0.0213)	(0.0256)
Observations	1,622,954	534,956
CADE tenders dropped	12,514	

Notes: All tenders involving any CADE-convicted firm are excluded. SE clustered at item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

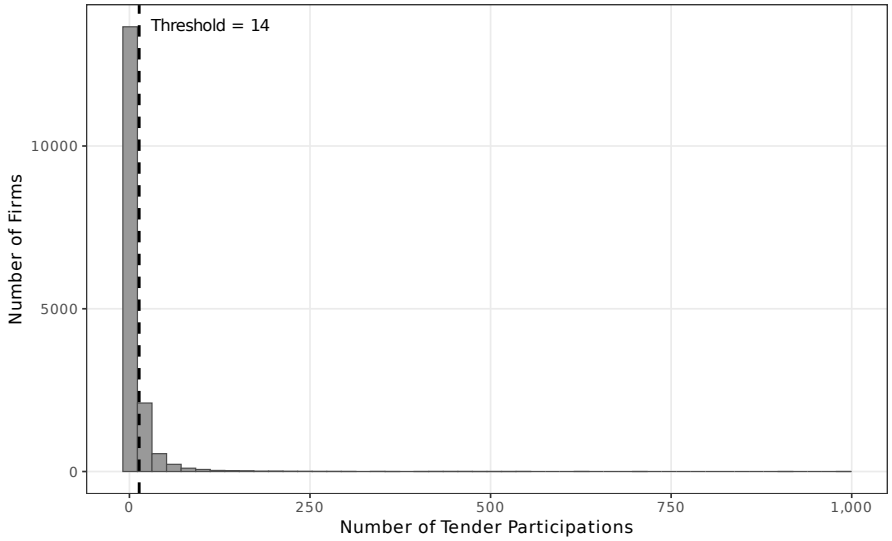

Figure B.2: Identification of the frequent-loser threshold via median + 1.5× IQR on the always-loser participation distribution. The cutoff yields 14 tender-items (13.5 statistical threshold) and flags 2,735 firms.

Appendix C. Identification Audits

Table C.1: Oster (2019) Coefficient Stability Bounds

	Full sample	Pregão	Convite
$\hat{\beta}_{\text{short}}$ (item + year FE)	0.0676	0.0954	0.0507
$\hat{\beta}_{\text{full}}$ (+ PBU FE)	0.0636	0.0933	0.0382
$\tilde{R}^2_{\text{short}}$	0.8790	0.8737	0.8857
$\tilde{R}^2_{\text{full}}$	0.8860	0.8821	0.8932
$R^2_{\max} = \min(1, 1.3\tilde{R}^2)$	1.0000	1.0000	1.0000
$\hat{\delta}$ (Oster bound)	261.64	634.00	43.62
$\beta^*(\delta = 1)$	-0.0005	0.0643	-0.1395
Observations	1,654,401	546,549	1,107,852

Notes: $\hat{\delta}$ is the degree of selection on unobservables (relative to observables) required to drive $\hat{\beta}$ to zero (Oster, 2019). $|\hat{\delta}| > 1$ indicates robustness: unobservables would need to be more important than observables to explain the coefficient. $\beta^*(\delta = 1)$ is the bias-adjusted coefficient assuming equal selection. $R^2_{\max} = 1.3\tilde{R}^2$ following Oster's recommended bound. Short model: item + year FE. Full model adds PBU FE (and convite control for full sample).

Appendix D. Threshold and Specification Robustness

Appendix E. Operational Metrics and Detector Comparison

Appendix F. Staggered Designs Attempted and Rejected

The difference-in-differences design exploiting Decreto 9.412/2018's 2018 raise of the convite cap fails the parallel-trends diagnostic at the available pre-treatment window.

Table C.2: Leakage Audit of the Item-Level Loser-Side Concentration AUC

Audit specification	What it tests	AUC	95% CI
Original (in-sample, full pool)	Reference: log(1+max tc) predicting any-cobidder participation.	0.995	$[0.995, 0.995]$
Audit 1: Different label, same score (scope)			
Same score → any-direct-CADE label	Whether the screen also flags items with direct CADE defendants (N+=54,039 items).	0.506	$[0.505, 0.507]$
Audit 2: 5-fold cross-validation at the cobidder-firm level (tautology)			
Pooled out-of-fold	Hold out $\sim 1/5$ of cobidders per fold; recompute tenders_count without their participation; predict items containing held-out cobidders using the recomputed score. Pool out-of-fold predictions across folds.	0.891	$[0.887, 0.894]$
Audit 3: Temporal holdout (generalization)			
Train 2009–2016 → test 2017–2019, any-cobidder	Compute tenders_count using only 2009–2016 items; predict items in 2017–2019.	0.864	$[0.859, 0.870]$
Train 2009–2016 → test 2017–2019, any-direct-CADE	Same temporal split, against direct-CADE label.	0.511	$[0.510, 0.513]$

Notes: The reference in-sample item-level AUC of 0.995 is partially tautological: cobidders are by construction always-loser firms with elevated participation counts, and an item that contains a cobidder participant therefore mechanically has an elevated max-tenders-count among its always-loser participants. Audit 2 breaks this construction by withholding cobidder firms from the score-construction step within each cross-validation fold and re-evaluating on items containing held-out cobidders. Audit 3 breaks the construction temporally: tenders-count is computed using only 2009–2016 participation, so the score for 2017–2019 items cannot directly encode their own cobidders. The screen retains discriminating power above 0.85 under both audits, indicating that the substantive predictive content is not fully tautological. The drop of approximately 0.10–0.13 AUC under audit is the pure-leakage component. Audits against direct-CADE labels return random AUC in all configurations, confirming the structural scope asymmetry documented in Table 4.

Table D.1: Price Coefficient Across IQR Multiplier Thresholds

Multiplier	Threshold	FL Firms	Coef.	SE	Observations
1.0 × IQR	10	3,442	0.0791***	(0.0227)	1,654,401
1.5 × IQR	14	2,735	0.0636***	(0.0215)	1,654,401
2.0 × IQR	17	2,093	0.0598***	(0.0221)	1,654,401
2.5 × IQR	20	1,778	0.0543**	(0.0228)	1,654,401
3.0 × IQR	24	1,456	0.0501**	(0.0237)	1,654,401

Notes: DV: log negotiated price. Each row re-estimates the baseline specification (item + year + PBU fixed effects) using a different IQR multiplier to define the FL threshold. The threshold equals median $+k\times$ IQR of the tenders-count distribution among always-losers. The baseline (1.5× IQR) is highlighted. SE clustered at item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table D.2: Alternative Clustering of Standard Errors

Clustering level	Coef.	SE	95% CI
Item level (baseline)	0.0636***	(0.0215)	[0.0214, 0.1058]
PBU level	0.0636***	(0.0144)	[0.0353, 0.0919]
Two-way (item + PBU)	0.0636***	(0.0240)	[0.0165, 0.1107]

Notes: DV: log negotiated price. All rows report the same baseline specification (item + year + PBU fixed effects) with different clustering of standard errors. The point estimate is invariant to clustering; only the standard errors and confidence intervals change. The two-way cluster uses the Cameron et al. (2011) multi-way approach. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table E.1: Operational Deployment Metrics: Precision, Recall, and Lift at Top-k

	In-sample (full 2009–2019)				Temporal holdout (score: 2009–2016)			
Top-k	TP	Precision	Recall	Lift	TP	Precision	Recall	Lift
50	15	0.300	0.078	26.2×	1	0.020	0.005	1.7×
100	17	0.170	0.088	14.8×	7	0.070	0.036	6.1×
250	40	0.160	0.207	14.0×	19	0.076	0.098	6.6×
500	66	0.132	0.342	11.5×	35	0.070	0.181	6.1×
1,000	97	0.097	0.503	8.5×	66	0.066	0.342	5.8×
Reference quantities								
Always-loser pool size N				16,843				
Cobidder positives N_+				193				
Base cobidder rate				0.0115				
ROC AUC (log(1+tc))		In-sample: 0.939				Temporal holdout: 0.864		

Notes: Score: $\log(1 + \text{tenders_count})$, ranked descending across the always-loser pool. Lift is precision divided by the base cobidder rate. The in-sample columns score firms using 2009–2019 participation counts; the temporal-holdout columns score using only 2009–2016 participation, evaluated against the same 193 cobidder labels. The temporal-holdout numbers are the operationally honest projection. The in-sample columns over-state precision by approximately 50% at top-500. We rely on the temporal-holdout figures in our operational claims; the in-sample column is reported for comparison and to make the leakage transparent. Computational cost on this run: roughly 1 CPU-second per state-year of award records on consumer-class hardware (Apple M1 / Ryzen 7 5800H equivalent). Operational deployment requires only contract-award records (winner identity, participant identity, item identifier); no bid microdata.

Table E.2: Incremental Value of Loser-Side Concentration Relative to the Full Imhof Benchmark

Model	Data requirements	AUC	95% CI	Δ vs Imhof full	DeLong p
Imhof full pipeline	All bid values within each tender	0.846	$[0.819, 0.873]$	+0.000	—
Binary FL flag	Winner + participants only	0.881	$[0.871, 0.892]$	+0.035	0.014
Continuous participation count	Winner + participants only	0.877	$[0.857, 0.898]$	+0.031	0.077
Imhof full + binary FL	Bid microdata + award records	0.942	$[0.927, 0.957]$	+0.096	< 0.001
Imhof full + participation count	Bid microdata + award records	0.944	$[0.929, 0.958]$	+0.098	< 0.001

Notes: All models are evaluated on the exact same always-loser subsample for which the full bid-distribution feature set is available. The relevant comparison is therefore not "does FL beat Imhof everywhere," but whether participation-only information remains informative when rich bid microdata are absent and whether it adds non-redundant signal when those microdata are present. The combined specifications answer the second question directly.

We report two attempted estimators alongside the failure mode. The Sun and Abraham (2021) interaction-weighted estimator yields imprecise coefficients with non-monotone dynamic patterns incompatible with the smooth treatment-onset narrative the cap raise would imply (Table F.1). A stacked difference-in-differences design constructed around the cap-raise event window returns coefficients that flip sign across stacks and lack a clean pre-period (Table F.2). We treat both estimators as null at the available bandwidths and report the empirical content of §7 as descriptive throughout (§10.1).

Table F.1: Staggered DiD: Callaway & Sant'Anna (Restricted Sample)

	C&S (2021)		TWFE	
Outcome	ATT	SE	Pre-trend	Post
Log Price	0.0143	(0.0396)	-0.0131	-0.0079
Log Firms (excl. FL)	-0.0220	(0.0333)	-0.0088	0.0113
Panel B: Statistical Power				
MDE (5%, two-sided)	0.0776			
MDE (80% power)	0.1109			
OLS benchmark	0.064			
Panel D: Sharp Entry Subsample				
C&S ATT	0.0143	(0.0389)		
Sharp entry markets	1,511 treated, 18,266 control			
Markets	19,777 (1,511 treated, 18,266 control)			
Sample restriction	≥3 pre & ≥3 post observations			

Notes: Panel A: C&S = Callaway & Sant'Anna (2021) doubly-robust estimator. Panel B: MDE = minimum detectable effect at stated significance/power levels. Panel C: Rambachan & Roth (2023) sensitivity bounds under smoothness restriction $\bar{M}$ on maximum change in the slope of the trend. Panel D: restricts treated markets to those with zero FL presence before treatment year.

References

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics 29*(2), 238–249.

Table F.2: Stacked DiD vs. Callaway & Sant'Anna

	Stacked (Cengiz et al.)		C&S (2021)	
Outcome	ATT	SE	ATT	SE
Log Price	-0.0059	(0.0138)	0.0143	(0.0405)
Log Firms (excl. FL)	0.0162*	(0.0084)	-0.0220	(0.0339)
Stacked obs.		715,116		
Cohorts		6		
Event window		±5 years		
Cohort × Market FE	YES		—	
Cohort × Year FE	YES		—	
Clustering		Market level		

Notes: Stacked regression following Cengiz et al. (2019). Each cohort-year sub-experiment includes treated markets entering in year g and all never-treated markets, within a symmetric ±5-year event window. Cohort-specific market and year fixed effects absorb level differences across sub-experiments. SE clustered at market level. C&S = Callaway & Sant'Anna (2021) doubly-robust estimator from Table F.1. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

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